

Experimental and numerical studies on the vortex-induced vibration of parallel twin steel box girders with varying gap distances and angles of attack

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ABSTRACT

Parallel twin steel box girders are increasingly favored by bridge designers because of their superior traffic capacity. However, the vortex-induced vibration (VIV) characteristics of twin steel box girders are more intricate than those of a single box girder owing to aerodynamic interference. In the present study, a systematic experimental investigation was conducted on the VIV characteristics of parallel twin steel box girders at varying gap distances and angles of attack (AoAs). The influence mechanism of aerodynamic interference on the VIV characteristics of twin box girders was investigated by a 2D static numerical simulation. Our investigation revealed that alternating vortex shedding between the windward and leeward box girders at both minimal and medium gap distances exerted an amplification effect on the VIV of twin box girders. At a small gap distance, the leeward box girder inhibited the shedding of vortices in the wake of the windward box girder, resulting in a reduction in VIV. At a large gap distance, the VIV characteristics of the windward box girder were very close to those of the single box girder; however, the leeward box girder was in a turbulent flow field formed by the wake of the windward box girder, thereby suppressing its VIV.

1. Introduction

Airflow passes through the bluff bridge section of a structure and separates from it, forming alternating shedding vortices. The proximity between the vortex shedding frequency of the wind flow and the self-oscillation frequency of the structure may induce the vortex-induced vibration (VIV) of the structure. Severe VIV may cause fatigue damage to bridge members and significantly impact traffic safety. Bridges with steel box girders tend to have low masses and damping ratios, making the main girder susceptible to VIV. Numerous steel box girders in bridge engineering projects constructed worldwide have encountered severe VIV occurrences [34,35,37,7]. To accommodate rising traffic volumes, modern bridges spanning rivers and seas require wide decks and additional lanes. Nevertheless, challenges in mechanics and architecture related to bridge aesthetics hinder the widening of the bridge deck. In recent years, designers of bridges have shown a preference for parallel twin steel box girders due to their superior traffic capacity, leading to the increased popularity of this type of bridge design.

Two main types of twin box girders exist: those with transverse cross beams and those without. Twin box girders with transverse cross beams are composed of two parallel longitudinal girders that are joined together by transverse cross beams. Several scholars have investigated the VIV [15,17,16,18,20,23,4,40,8], flutter stability [30,31,33,38,39,41,44], and aerodynamic force [1,19,25,3,36,39,42,43,45,6] of twin box girders with transverse cross beams using numerical simulations and stationary or dynamic wind tunnel tests. These scholars have identified the gap distance and angle of attack (AoA) as crucial factors that affect the characteristics of VIV, flutter stability, and aerodynamic force of twin box girders.

Twin box girders without transverse cross beams can be categorized into two scenarios. The first scenario is a new bridge running parallel to an existing bridge, such as the Tacoma Bridge in the USA [11], the Jindo Bridge in Korea [13,28,29,32], the Haihe Bridge in China [27] and the Ewijk Bridge in the Netherlands [2]. Sometimes, long-span bridges may be directly built with parallel twin girders, such as the Fred Hartman Bridge in the USA [26], the Hongdao Bridge [21], and the Pingsheng Bridge in China [5]. Parallel twin box girders without transverse cross

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Nomenclature

A	Vibration amplitude
B	Breadth of single box girder model
D	Net gap distance between twin girder models
D/B	Gap distance ratio
H	Girder model height
L	Section model length
m	Mass per unit length
U	Incoming wind speed
Ur	Reduced wind velocity
f_h	Vertical natural frequency
ζ_h	Vertical damping ratio
h	Allowable vertical VIV amplitude
γ	Partial factor for VIV
$y+$	Dimensionless wall distance
α	Angle of attack (AoA)

beams constructed during different periods typically share the following characteristics: (i) profiles of the two girders are typically similar but may differ; (ii) the gap distance between the two girders is relatively large; and (iii) the two bridges typically do not share a bridge tower. In contrast, twin box girders constructed simultaneously often share the following features: (i) both box girders tend to have similar geometries; (ii) the gap distance between the two girders is kept relatively small to reduce overall construction costs; and (iii) two girders typically share bridge towers or have towers connected to enhance the lateral stiffness of the bridge structure and effectively resist lateral loads acting upon it.

To examine the VIV [13,14,2,24,27,28,29,32,5], flutter stability [5], and aerodynamic force [10,11,2,21,5,9] characteristics of twin box girders without transverse cross beams, wind tunnel tests, field observations, and computational fluid dynamics (CFD) have been used.

The aerodynamic characteristics of twin box girders are more complicated than those of a single box girder owing to aerodynamic interference between the windward and leeward box girders. Thus, several studies have been conducted to uncover the aerodynamic interference law on the VIV of parallel twin box girders. Wind tunnel tests were conducted by Chen et al. [5] on sectional models of twin box girders—the Ping Sheng Bridge and the Hongdao Bridge, in China—which revealed that the aerodynamic interference effect between the two girders cannot be ignored. This effect may cause an increase in VIV amplitudes, and girders with different gap distances may experience varying levels of aerodynamic interference. The interference VIV between two parallel cable-stayed bridges and the mutual motion of both leeward and windward girders were investigated by Kim et al. [13]. According to the parametric wind tunnel tests, interference VIV can occur. Seo et al. [32] detected the interference effect on VIV in both field observations and wind tunnel tests conducted on twin cable-stayed bridges. The VIV of the windward girder is amplified by the interference effect between the parallel girders.

Several researchers have investigated how the gap distance affects the VIV characteristics of twin box girders. Kimura et al. [14] investigated the wind-induced response characteristics of two parallel girders with varied sectional geometries at different gap distances using elastically supported sectional models. The authors found that the response characteristics are complex and vary with changing gap distance. Even at gap distances reaching eight times the girder breadth, parallel girders can have substantial effects. According to Park et al. [29], gap distances reaching five to seven times the girder depth significantly impact the interactive VIV in two parallel cable-stayed bridges. This critical gap distance is closely linked to the travel distance of one vortex during a single period of girder oscillation.

Several scholars have examined the effects of AoA on the VIV

characteristics of parallel twin box girders. An experimental investigation was conducted by Liu et al. [24] to study the aerodynamic performance of long-span rail-cum-road bridges with parallel twin box girders. AoAs were found to affect the vertical and torsional VIV amplitudes of highway and railway girders. AoA should be considered when evaluating the influences of aerodynamic measures on VIV suppression for parallel twin girders. According to Meng et al. [27], the VIV amplitudes of the windward girder are often fairly close to those of the single girder. It was also shown that the AoA affects the VIV responses of two adjacent girders.

The literature suggests that gap distance and AoA are significant factors that influence the VIV characteristics of parallel twin box girders. Although many scholars worldwide have studied the VIV characteristics of parallel twin box girders and obtained some findings, it is problematic to accurately and systematically identify the impacts of gap distance and AoA on the VIV characteristics of parallel twin box girders based on previous research due to the complexity. It is necessary to conduct additional research on this issue. In this paper, a series of sectional model vibration tests were conducted on parallel twin steel box girders in a wind tunnel within the context of the Hongdao Channel Bridge in China. Initially, we conducted a detailed analysis of the VIV characteristics of parallel twin box girders for sixteen different gap distances at a large AoA of $+5^\circ$. Then, we examined the VIV characteristics of parallel twin box girders at eight different AoAs ranging from -5° to $+5^\circ$ for four representative gap distances. A comparison was made between these VIV characteristics and those obtained from the single box girder. Finally, the CFD simulation approach was applied to visualize the flow field. In this study, numerical simulations of 2D stationary models were employed to provide preliminary discussions and explanations of the mechanism of aerodynamic interference in wind tunnel sectional model vibration tests. This paper is organized as follows: Section 2 introduces the engineering background, the design of the sectional models, and the configuration of the wind-tunnel tests. Section 3 presents the experimental results, including the VIV responses of the single box girder under varying AoAs and those of the twin box girders under different AoAs and spacing ratios. Section 4 details the numerical simulations and flow-field analyses conducted to elucidate the underlying aerodynamic interference mechanisms. Section 5 synthesizes the main findings and draws the overall conclusions. To provide a clearer overview of the research framework, the overall experimental and numerical workflow is illustrated in Fig. 1.

2. Experimental introduction of the VIV of parallel twin box girders

2.1. Engineering background

The Qingdao Bay Bridge stretches across the waters of Jiaozhou Bay in China, and the Hongdao Channel Bridge is one of the three channel bridges within this bay. As shown in Fig. 2, the Hongdao Channel Bridge with a span of (60 +120 +120 +60) m consists of two parallel steel box girders supported by a single tower. The girders of the bridge are made from steel box stiffeners, and the towers are solid concrete columns with octagonal shapes that vary in sections. The geometric parameters of the twin steel box girders for the Hongdao Channel Bridge are shown in Fig. 3, with a breadth of 18.5 m, a height of 3.2 m for each girder, and a gap distance of 12 m.

2.2. Sectional model and experimental setup

Sectional model vibration tests of parallel twin box girders were carried out at a geometric scale of 1:50. The tests were based on the Hongdao Channel Bridge, and they disregarded barriers and transverse slopes. Fig. 4 presents the geometric parameters of the parallel twin box girder models. The single box girder model had a breadth (B) of 370 mm, a height (H) of 64 mm, and a length (L) of 2000 mm. The gap

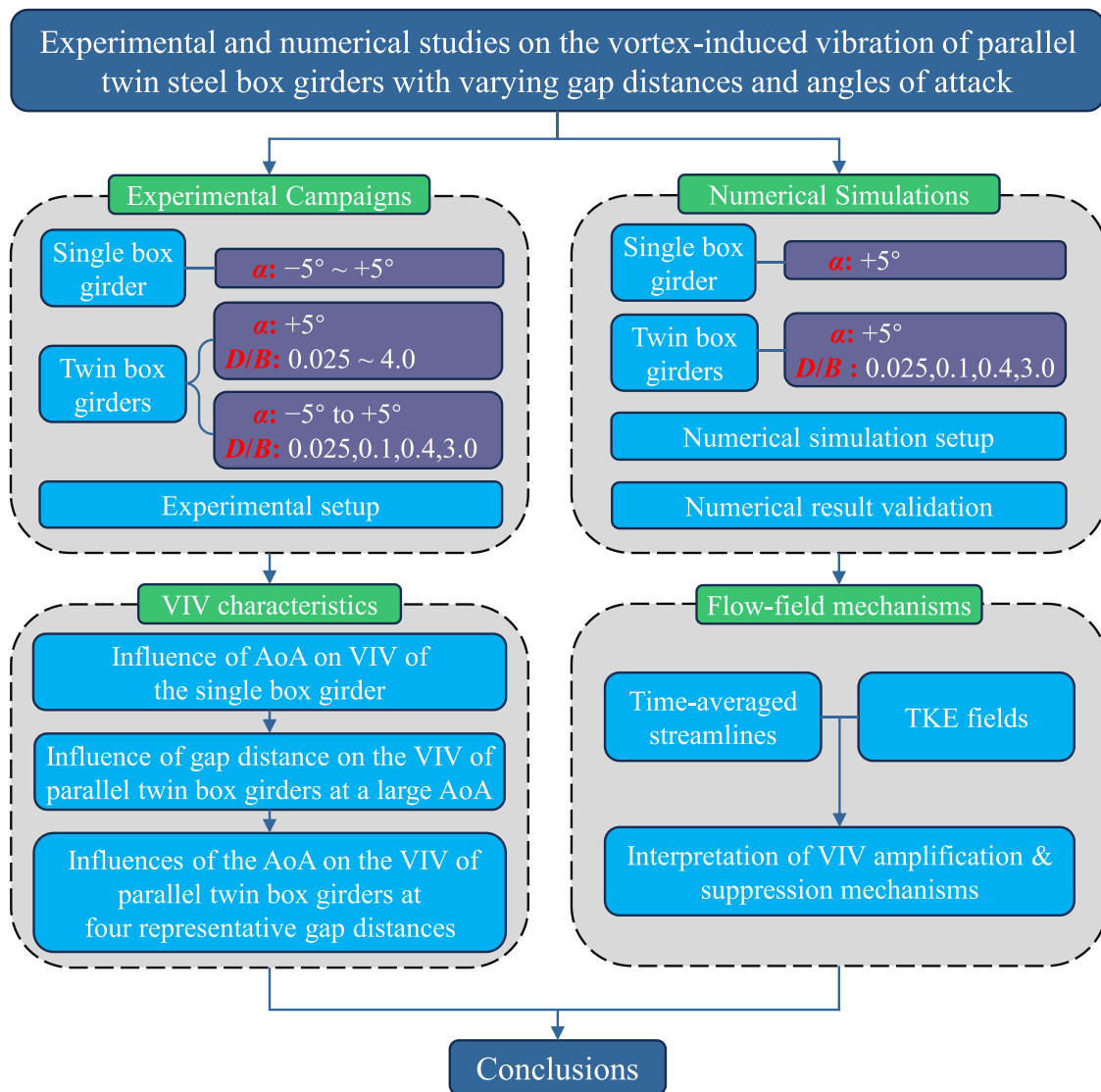


Fig. 1. Experimental and numerical workflow.



Fig. 2. Photograph of the Hongdao Channel Bridge, China.

distance between the twin box girders was defined as D . The AoA was represented by the symbol α . The positive AoA of incoming wind was defined as the oblique upward flow toward the model. The negative AoA of incoming wind was defined as the oblique downward flow toward the model. The Hongdao Channel Bridge experienced a maximum AoA of $\pm 3^\circ$. However, to increase the applicability of this study to practical engineering, the maximum AoA considered in this study was $\pm 5^\circ$. This adjustment was made to account for areas with highly pronounced AoA effects of the wind field, such as mountainous regions.

The tests were performed in the low-speed test section of the STDU-1 wind tunnel at Shijiazhuang Tiedao University. This test section was 4.4 m in width, 3 m in height, and 24 m in length, allowing for continuous regulation of the wind speed from 1 m/s to 30 m/s. The experiment was conducted in a uniform flow field, and the longitudinal

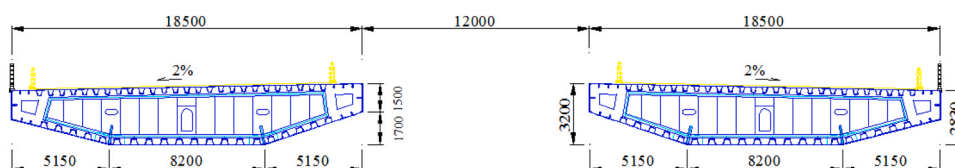


Fig. 3. Geometry parameters of the twin box girders of the Hongdao Channel Bridge, China (unit: mm).

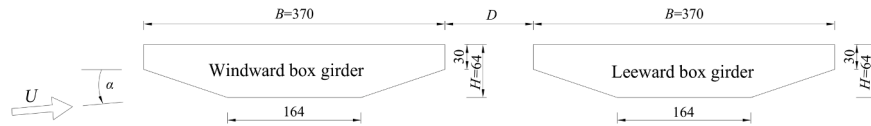


Fig. 4. Geometric parameters of the parallel twin box girder models (unit: mm).

turbulence intensity in the empty wind tunnel was approximately 0.2 %. The maximum blockage ratio of the box girder model was 3.2 %, which is well below the commonly accepted 5 % threshold, indicating that the influence of model blockage on the VIV characteristics of the twin box girders can be considered negligible.

Fig. 5 presents the spring-suspended sectional model system, designed to allow flexible adjustment of key parameters such as gap distance and AoA for the box girders. This design was necessary given the complexity of the VIV tests on twin box girder models. Two girder models, made of ABS plastic and stiffened by a central steel shaft, were each suspended independently by eight springs. The upper tension springs connected to sliders in the top slide rails fixed to the wind tunnel's low-speed test section via turnbuckles, while the lower tension springs were connected to turnbuckles through force transducers, which in turn linked to sliders in the bottom slide rails.

The spacing between the two girders was adjusted by moving the sliders along the rails, and the rails were arranged to prevent collisions between components. Both windward and leeward girders could be rotated synchronously around the steel shaft axis to set identical AoAs, then locked in place. Vertical alignment was achieved using the turnbuckles so that the upper surfaces of both models lay in the same plane, enabling AoA variation at different gap distances.

The force transducers used in this study, manufactured by the China Academy of Aerospace Aerodynamics Technology, have a measuring range of 0–100 kg, an accuracy level of 0.05, and a sampling frequency of 500 Hz. The displacement response was obtained by dividing the measured force by the calibrated spring stiffness. All transducers were calibrated prior to the experiments to ensure reliable measurements. Dimensions of the girders and their spacing were measured with a vernier caliper and a ruler, with accuracies of ± 0.2 mm and ± 0.5 mm, respectively.

The dynamic parameters of the models in the spring-suspended sectional model system are presented in Table 1. The two box girder models possessed comparable masses (m), vertical natural frequencies (f_h), and damping ratios (ζ_h). The first torsional frequency of the actual bridge was 2.27 Hz, which was almost three times its vertical frequency. The probability of triggering VIV in the torsional degree of freedom was approximately negligible due to the high torsional natural frequency. Furthermore, VIV is a type of vibration occurring in a single mode that is not coupled with the vertical or bending modes. Consequently, only the parameters related to the vertical bending mode were relevant in this

Table 1

Dynamic parameters of the two box girder models.

Parameter	Full-scale bridge	Windward box girder model	Leeward box girder model
Geometric scale ratio	—	1:50	1:50
Mass m (kg/m)	23250	9.30	9.30
Vertical natural frequency f_h (Hz)	0.77	5.74	5.77
Vertical damping ratio ζ_h (%)	0.50	0.52	0.55

study.

According to the *Wind-resistant Design Specification for Highway Bridges* (China) [12], the allowable vertical VIV amplitude is defined as

$$h = \gamma \times \frac{0.04}{f_h} \quad (1)$$

where h is the allowable amplitude, and γ is the partial factor for VIV. Since h in this study is obtained from wind tunnel measurements, γ is taken as 1. Based on this equation, the allowable vertical VIV amplitude for the actual bridge is calculated to be 0.052 m, which corresponds to 1.04 mm in the model scale. For clarity and to facilitate comparison between the measured VIV amplitudes and the allowable value specified in the standard, a black horizontal line representing the allowable vertical VIV amplitude has been added to all figures presenting VIV amplitude results.

3. Experimental results

3.1. Influence of AoA on VIV of the single box girder

Sectional model vibration tests of the single box girder were first conducted, using AoAs of $+5^\circ$, $+4^\circ$, $+3^\circ$, $+2^\circ$, $+1^\circ$, 0° , -3° , and -5° . VIVs were detected at AoAs of $+5^\circ$, $+4^\circ$, and $+3^\circ$. Preliminary tests, in which the VIV amplitudes were assessed only through visual observation rather than by precise instrumental measurements, revealed that significant VIVs occurred for both the single box girder and the parallel twin box girders at positive AoAs, whereas almost no VIV was observed at negative AoAs. Based on these observations, the positive AoAs were set with a finer resolution, while only two negative AoAs (-3° and -5°) were selected.

Notably, the displacement of the models was described using reduced velocity and dimensionless amplitude. The reduced velocity U_r was defined as $U/(f_h B)$, with U representing the incoming wind speed. The dimensionless amplitude was denoted by $100 \bullet (A/H)$, with A representing the amplitude of vibration.

Fig. 6 displays the VIV responses for the single box girder with a vertical natural frequency f_h of 5.74 Hz at different AoAs. It can be observed that, for AoAs of $+4^\circ$ and $+5^\circ$, the VIV amplitudes exceeded allowable amplitude specified by the code, whereas for the other tested AoAs the amplitudes remained below the allowable limit. The reduced velocity ranged from 1.8 to 2.7 in the VIV lock-in region for the single box girder at $+5^\circ$ AoA, and the maximum dimensionless amplitude was achieved at a reduced wind speed of 2.6, which was approximately 3.8. As the AoA decreased, the length of the lock-in region of the single box girder, the maximum amplitude, and the reduced wind speed related to the maximum amplitude all decreased. This trend implies that the VIV

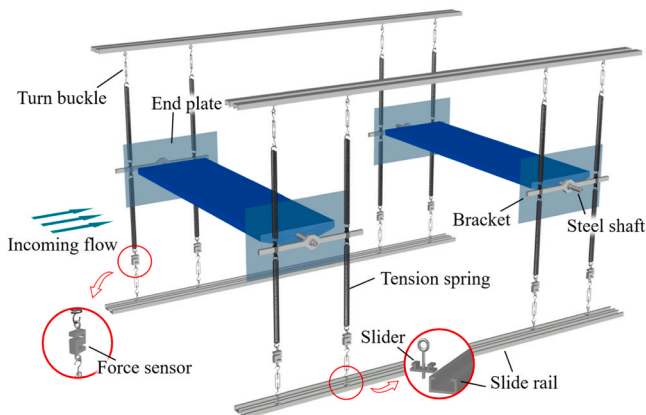


Fig. 5. Assembly of the spring-suspended sectional model system.

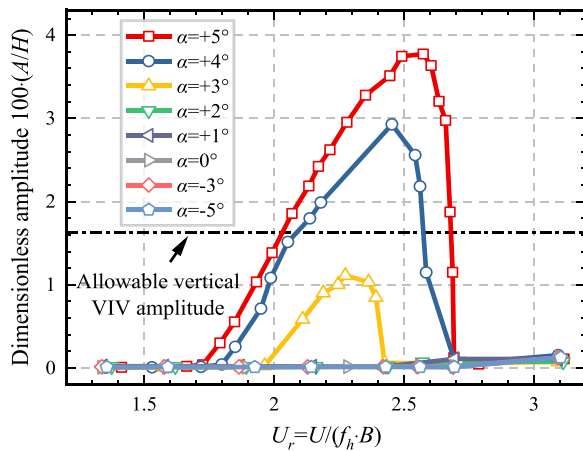


Fig. 6. VIV responses for the single box girder at different AoAs.

performance of the single box girder degrades at $+5^\circ$ AoA and improves gradually as the AoA decreases.

3.2. Influence of gap distance on the VIV of parallel twin box girders at a large AoA

Sectional model vibration tests were conducted on parallel twin box girders at an AoA of $+5^\circ$, as the poorest VIV performance was observed for a single box girder at this value. Additionally, the length of the lock-in region was the widest. To comprehensively investigate the influence of spacing on the VIV characteristics of parallel twin box girders, the spacing ratio D/B —defined as the ratio of the net gap distance D between the girders to the breadth B of a single box girder—was selected based on the actual spacing range in the bridge projects and appropriately extended beyond this practical range. Sixteen values of D/B were chosen to cover this extended range and to thoroughly capture potential aerodynamic interference effects. These values are: 0.025, 0.05, 0.075, 0.1, 0.15, 0.2, 0.25, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0, 2.5, 3.0, and 4.0.

Fig. 7 illustrates the VIV responses of the single box girder and the windward box girder in a parallel twin box girder system with varying

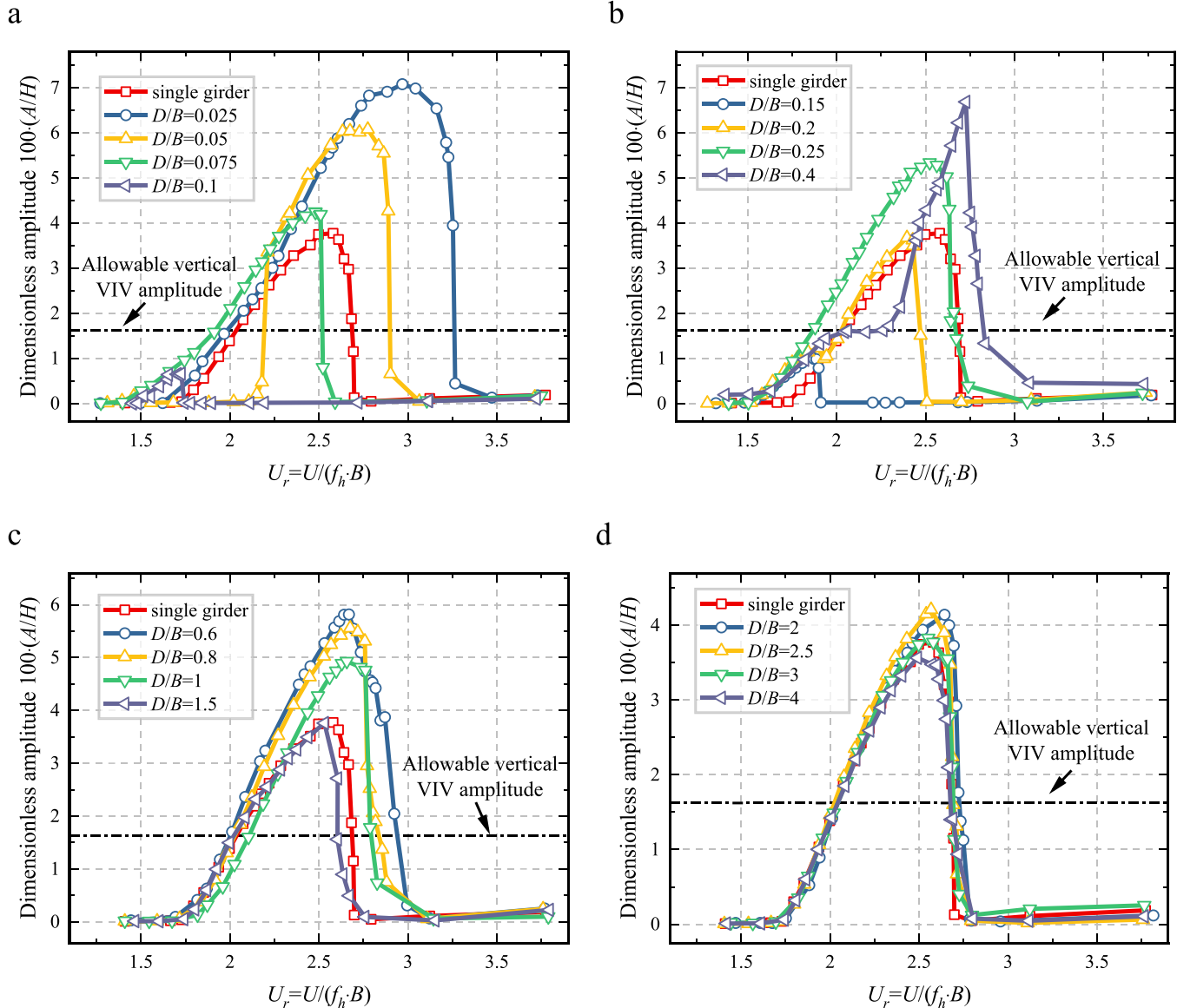


Fig. 7. VIV responses for the single box girder and windward box girder at $+5^\circ$ AoA in gap distances of (a) $0.025 \leq D/B \leq 0.1$, (b) $0.15 \leq D/B \leq 0.4$, (c) $0.6 \leq D/B \leq 1.5$ and (d) $2.0 \leq D/B \leq 4.0$.

gap distances, while Fig. 8 demonstrates the VIV responses for the single box girder and the leeward box girder in a parallel twin box girder system with different gap distances. The results of sixteen gap distances were classified into four groups for convenient observation of the variations in VIV amplitudes in the windward and leeward box girders. The results from each group with four gap distances were compared against the results from the single box girder. It can be observed that, the VIV amplitudes of both the windward and leeward girders generally exceed the allowable vertical VIV amplitude prescribed by the code under most spacing ratios. Fig. 9 illustrate the maximum amplitude of VIV for the windward and leeward box girders, respectively, at different D/B ratios.

An integrated analysis of Figs. 7 and 9 revealed that the effect of gap distance on the VIV characteristics of the windward box girder at $+5^\circ$ AoA could be grouped into the following four intervals:

For $0.025 < D/B < 0.075$, the maximum VIV amplitude of the windward box girder was significantly greater than that of the single box girder. The amplitude of VIV increased as the gap distance decreases. The reduced velocity for the onset of VIV of the windward box girder was proximate to that of the single box girder. Nevertheless, at the maximum VIV amplitude and the end of the lock-in region, the required

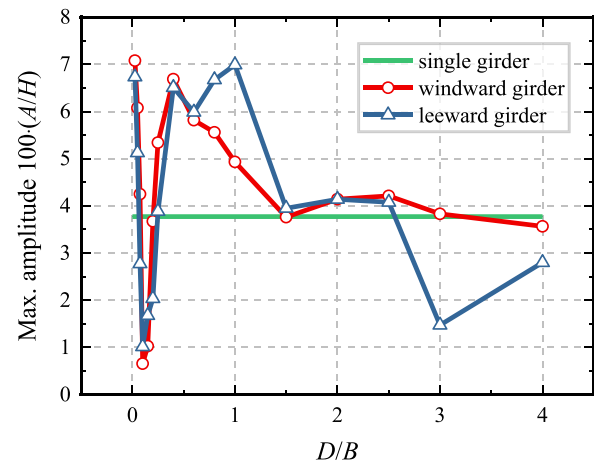


Fig. 9. Maximum VIV amplitudes with D/B of twin box girders.

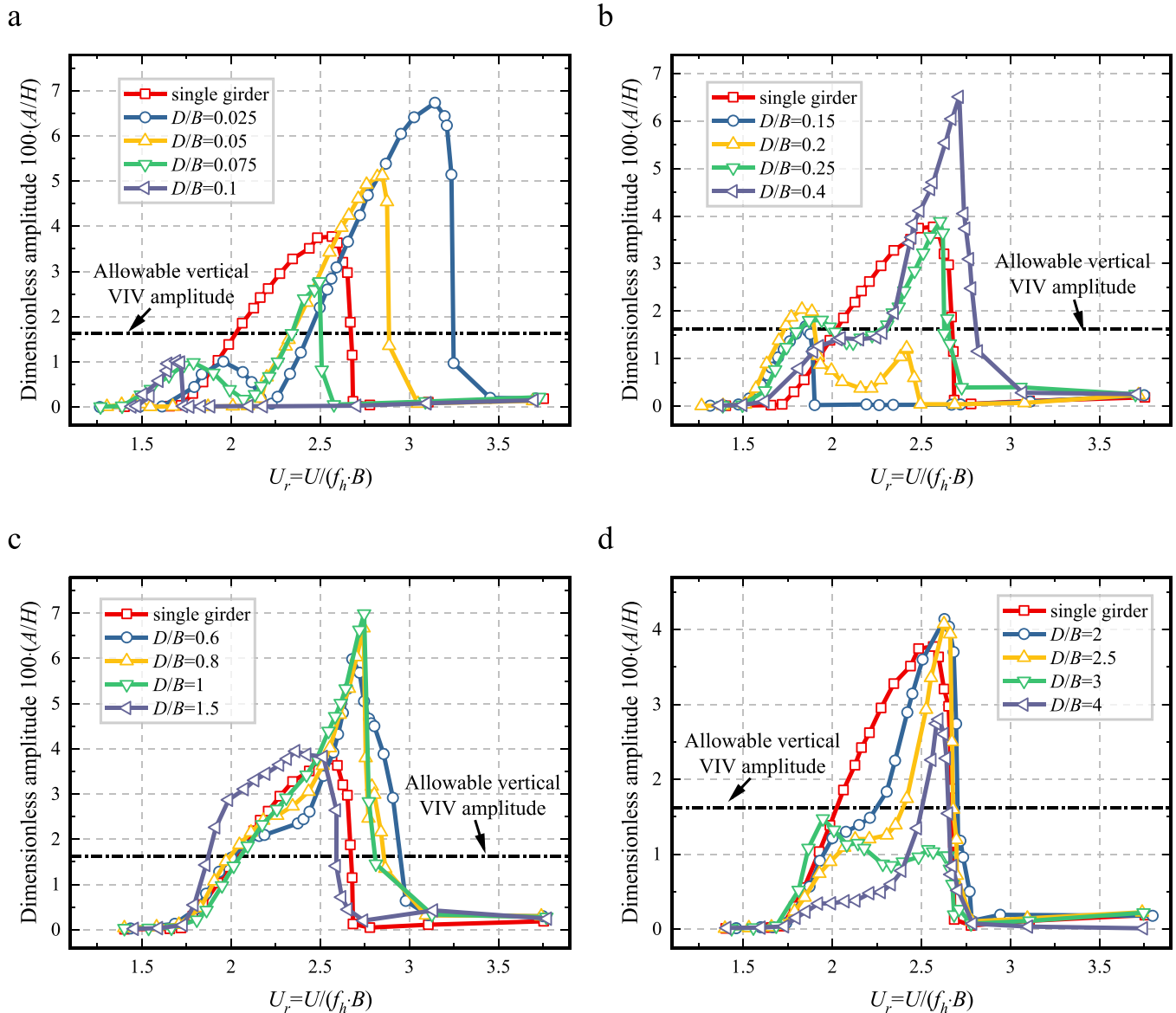


Fig. 8. VIV responses of the single box girder and leeward box girder at $+5^\circ$ AoA in gap distances of (a) $0.025 \leq D/B \leq 0.1$, (b) $0.15 \leq D/B \leq 0.4$, (c) $0.6 \leq D/B \leq 1.5$ and (d) $2.0 \leq D/B \leq 4.0$.

reduced wind speeds were higher for the windward box girder than for the single box girder.

For $0.075 < D/B < 0.2$, the maximum amplitude of the windward box girder was smaller than that of the single box girder. The onset reduced velocity of VIV for the windward box girder approached that of the single box girder, whereas the reduced wind speeds at the maximum amplitude and the end of the lock-in region were significantly lower than those of the single box girder.

For $0.2 < D/B < 1.5$, the maximum amplitude of the windward box girder first increased and then decreased. Moreover, the maximum amplitude under each gap distance was greater than that of the single box girder with the gap distance. The maximum amplitude of the windward box girder was approximately 1.7 times higher than that of the single box girder at a D/B of 0.4. For the windward box girder, the reduced velocities at the beginning and maximum amplitude of the lock-in region were close to those of the single box girder. However, the reduced wind speed at the end of the lock-in region was slightly higher for the windward box girder than for the single box girder.

For $1.5 < D/B < 4.0$, the maximum dimensionless amplitude and reduced velocities at the beginning and end of the lock-in region for the windward box girder approached those of the single box girder. Additionally, the reduced velocity at the maximum amplitude of the windward box girder also approached that of the single box girder.

An integrated analysis of Figs. 8 and 9 revealed that the VIV performance of the leeward box girder at a $+5^\circ$ AoA was affected by the gap distance, which could further be segregated into four intervals:

For $0.025 < D/B < 0.075$, the maximum VIV amplitude of the leeward box girder was significantly higher than that of the single box girder. Additionally, the amplitude increased as the gap distance decreased. The onset velocity of VIV for the leeward box girder was proximate to that of the single box girder. However, both the reduced velocity at the maximum dimensionless amplitude and the end of the lock-in region were higher for the leeward box girder than for the single box girder.

For $0.075 < D/B < 0.25$, the maximum amplitude of the leeward box girder is notably smaller than that of the single box girder. While the onset reduced velocity of VIV is proximate for both the leeward box girder and the single box girder, the reduced velocity at the maximum amplitude and the end of the lock-in region is considerably lower for the former.

For $0.25 < D/B < 2.5$, the maximum amplitude of the leeward box girder first increases and then decreases but remains greater than that of the single box girder for each increase in gap distance. At a D/B value of 1.0, the maximum amplitude of the leeward box girder is approximately 1.8 times greater than that of the single box girder. The reduced wind speeds at the beginning of the lock-in region and at the maximum amplitude of the leeward box girder are similar to those of the single box girder, although the end reduced velocity of the VIV is slightly increased.

For $2.5 < D/B < 4.0$, the reduced velocity at the maximum amplitude and at the beginning and end of the lock-in region for the leeward box girder closely resemble those of the single box girder. However, the maximum dimensionless amplitude of the leeward girder is smaller than that of the single girder.

3.3. Influences of the AoA on the VIV of parallel twin box girders at four representative gap distances

According to the previous section, the effect of the gap distance on both windward and leeward box girders at $+5^\circ$ AoA could be divided into four intervals. To investigate the VIV characteristics of parallel twin box girders at different AoAs, the following four representative gap distance ratios were selected from the mentioned intervals: $D/B = 0.025$ (minimal gap distance), $D/B = 0.1$ (small gap distance), $D/B = 0.4$ (medium gap distance), and $D/B = 3.0$ (large gap distance). In the VIV test, the parallel twin box girder models shared the same ranges of AoAs and reduced velocities as the single box girder model.

The wind tunnel tests revealed that VIV is prominent in the twin box girder models at gap distances of $D/B = 0.025$ and $D/B = 3.0$ and at AoAs of $+5^\circ$, $+4^\circ$, and $+3^\circ$. VIV also occurs at gap distances of $D/B = 0.1$ and $D/B = 0.4$ and at AoAs of $+5^\circ$, $+4^\circ$, $+3^\circ$, and $+2^\circ$.

Figs. 10 and 11 display the VIV responses of various AoAs at four gap distances. One figure depicts the windward box girder, while the other shows the leeward box girder. For the windward girder, the VIV amplitudes at certain positive AoAs exceed the allowable vertical VIV amplitude across all four representative spacing ratios. In contrast, for the leeward girder, exceedances of the allowable vertical VIV amplitude are observed only at specific positive AoAs when the spacing ratios are 0.025 and 0.4. Fig. 12 presents a comparison of the maximum amplitude curves for the VIV of the windward and leeward box girders with those of the single box girder for a range of AoAs. After reviewing Figs. 6, 10, 11, and 12 extensively, the following observations are evident:

For D/B values equal to 0.025 and AoA values between $+3^\circ$ and $+5^\circ$, the amplitudes of both the windward and leeward box girders and the length of the lock-in region all increase. The maximum amplitudes of both the windward and leeward box girders are greater than those of the single box girder. While the onset reduced velocities of VIV of both the windward and leeward box girders are similar to those of the single box girder, the end reduced velocities of VIV for both girders are increased. This increase indicates that aerodynamic interference amplifies the VIV of twin box girders with minimal gap distances of $D/B \approx 0.025$, increasing the amplitude and lock-in region length.

For D/B values equal to 0.1 and AoA values of $+5^\circ$ and $+4^\circ$, the maximum amplitudes of both the windward and leeward box girders are smaller than the maximum amplitude of the single box girder. Additionally, the lengths of the lock-in region for the windward and leeward box girders are shorter than that of the single box girder. However, the difference in the maximum amplitudes and lock-in region lengths between twin box girders and the single box girder at $+4^\circ$ AoA is less than at $+5^\circ$ AoA. At an AoA of $+3^\circ$, the windward box girder exhibits a larger maximum amplitude than the single box girder. Although the maximum amplitude of the leeward box girder is smaller than that of the single box girder, the difference further decreases relative to AoA values of $+5^\circ$ and $+4^\circ$. The single box girder does not exhibit any VIV. In contrast, both the windward and leeward box girders display relatively obvious instances of VIV at an AoA of $+2^\circ$. This finding shows that twin box girder VIV characteristics at small gap distances of $D/B \approx 0.1$ are related to the AoA. As the AoA decreases, the influences of aerodynamic interference on the VIV characteristics of twin box girders gradually change from reduction to amplification.

At a D/B of 0.4, from $+2^\circ$ to $+5^\circ$ AoA, the amplitudes of the windward and leeward box girders increase overall, and the length of the lock-in region increases. The maximum amplitudes of both windward and leeward box girders at $+5^\circ$, $+4^\circ$, and $+3^\circ$ AoAs are greater than that of the single box girder. The onset reduced velocities of VIV for both windward and leeward girders are quite close to the single girder, but the end reduced velocities of the former systems are slightly higher than the latter. At $+2^\circ$ AoA, there is no distinct lock-in region for the single box girder, while distinct lock-in regions can be observed for both windward and leeward box girders. The results indicate that at medium gap distances of $D/B \approx 0.4$, aerodynamic interference amplifies the VIV of twin box girders. This phenomenon is evidenced by the increases in the dimensionless amplitude and the length of the lock-in region. The trend of the windward box girder amplitude with reduced velocity at $+5^\circ$ AoA differs slightly from that at other AoAs. This divergence may be attributed to the relatively significant interference effect of the leeward box girder on the VIV of the windward box girder at this AoA (Fig. 10(c)). The trends of the windward and leeward box girder amplitudes with reduced velocity at $+5^\circ$ AoA are usually consistent, as presented in Figs. 10(c) and 11(c).

At a D/B of 3.0, from $+3^\circ$ to $+5^\circ$ AoA, the amplitudes of both the windward and leeward box girders increase overall, as does the length of the lock-in region. Although the maximum amplitude of the windward

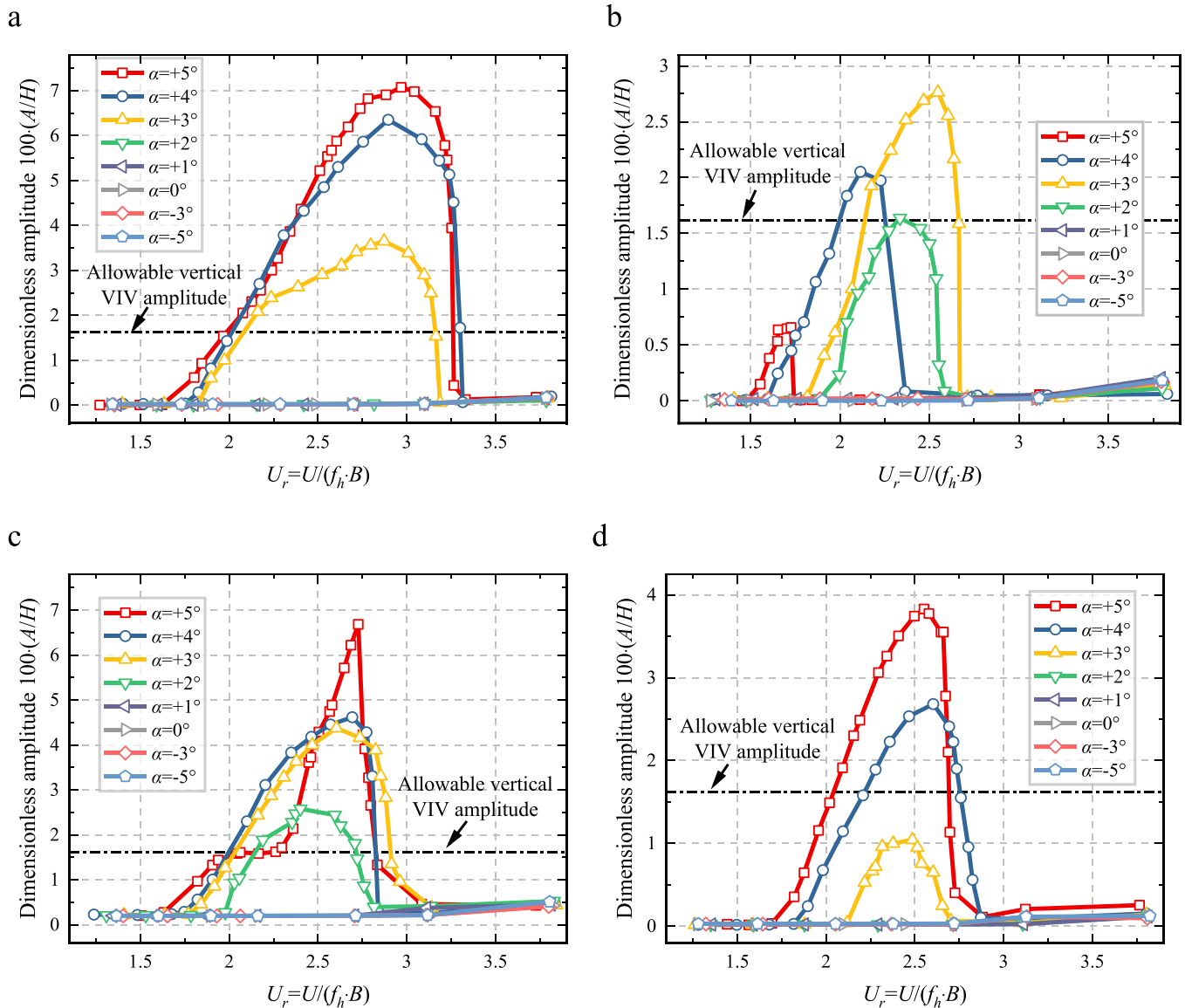


Fig. 10. VIV responses of the windward box girder at different AoAs and at gap distances of (a) $D/B = 0.025$, (b) $D/B = 0.1$, (c) $D/B = 0.4$ and (d) $D/B = 3.0$.

box girder is almost the same as that of the single box girder, the maximum amplitude of the leeward box girder is smaller than that of the single box girder. The windward and leeward box girders have similar onset and end reduced velocities of VIV relative to the single box girder. These findings suggest that the presence of the windward box girder reduces the VIV of the leeward box girder at large gap distances of $D/B \approx 3.0$; the interference of the leeward box girder on the VIV of the windward box girder is negligible.

According to the separate images presented in Fig. 12, when the gap distance D/B is 0.025 or 0.4, the aerodynamic interference exerts an amplification effect on the VIV of both the windward and leeward box girders relative to the maximum VIV amplitude of the single box girder at each AoA. When D/B reaches 3.0, the VIV of the windward box girder becomes very similar to that of a single box girder at various AoAs. Moreover, the aerodynamic interference has a reduction effect on the amplitude of the VIV of the leeward box girder. Unlike the other three gap distances, the effects of aerodynamic interference on the VIV of the windward and leeward box girders are not consistent with changing AoAs when D/B equals 0.1. Specifically, the VIV of both girders first decreases and then amplifies as the AoA decreases from $+5^\circ$ to $+1^\circ$. This discrepancy in VIV characteristics at the same gap distance relative

to the AoA indicates that there is a difference in the range of the four gap distance intervals classified according to the effect (amplification or reduction) of aerodynamic interference on VIV at different AoAs.

4. Numerical investigation of parallel twin box girders

4.1. Numerical simulation setup

The VIV characteristics of parallel twin box girders are affected by the gap distance and AoA, in essence, through the change in the surrounding flow field. These changes in the flow field cannot be fully understood using wind tunnel tests alone. Hence, it is necessary to employ numerical simulation techniques to analyze the effects of the variations in gap distance and AoA on VIV characteristics from the perspective of the flow field. The two-dimensional static CFD method for single box girders and twin box girders with different gap distances can simulate the flow field around stationary box girder models to provide a preliminary analysis of the mechanism of the aerodynamic interference of VIV in wind tunnel sectional model vibration tests.

The VIV characteristics of twin box girders with a $+5^\circ$ AoA are significantly impacted by aerodynamic interference, resulting in more

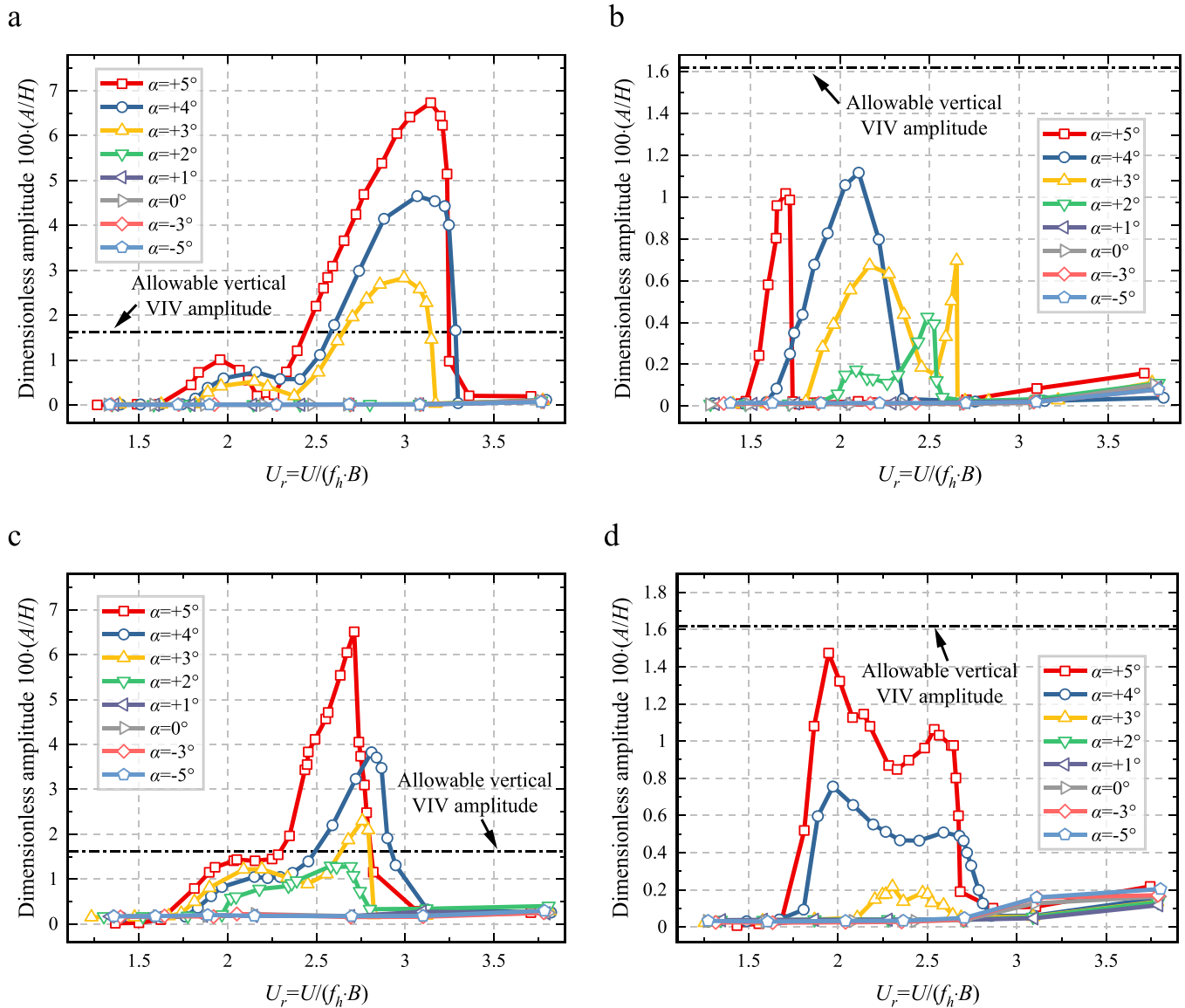


Fig. 11. VIV responses of the leeward box girder at different AoAs and at gap distances of (a) $D/B = 0.025$, (b) $D/B = 0.1$, (c) $D/B = 0.4$ and (d) $D/B = 3.0$.

pronounced amplification or suppression effects of VIV than other AoAs. Therefore, the $+5^\circ$ AoA is a highly representative choice for investigating the effects of interference on the VIV characteristics of twin box girders. Numerical simulations were conducted for the following five cases: a reference case using a single box girder and four variations with twin box girders with different gap distances, namely, minimal ($D/B = 0.025$), small ($D/B = 0.1$), medium ($D/B = 0.4$), and large ($D/B = 3.0$).

As shown in Fig. 13(a), a 2D computational domain was used with boundary conditions for the flow field in the numerical simulation. The input wind speed at the velocity inlet was 10 m/s. The maximum blockage ratio in the numerical simulation was 2.35 %, which is also well below the commonly accepted 5 % threshold, ensuring that the blockage effect on the computed flow field is negligible. To discretize this computational domain, a structured mesh was utilized and suitably refined at the corners of the model where the flow field changes were increasingly complex. Fig. 13(b) comprehensively displays the mesh generation characteristics in the local amplifications of the single box girder and twin box girders with varying gap distances. The origin of the coordinate system is located at the center of the windward box girder. The distances between the X-axis and the upper and lower boundaries are both $10B$, while the distances between the Y-axis and the inlet and

outlet boundaries are $10B$ and $20B$, respectively.

We conducted numerical simulations using the commercial software ANSYS Fluent with the $k-\omega$ SST turbulence model to simulate the flow fields around the single box girder and twin box girders situated at varying gap distances. The pressure-velocity coupling equations were solved by the SIMPLEC scheme. The gradient and pressure simulations of spatial discretization used least squares cell-based and second order programs, respectively, whereas the momentum, turbulent kinetic energy (TKE), and specific dissipation rate simulations adopted a second-order upwind program. Steady-state calculations were performed first. Once the calculations were stable, we conducted transient calculations with the second-order implicit scheme in the transient formulation.

The majority of y^+ values nearest to the wall were controlled below 1, ensuring that the flow separation phenomena could be accurately simulated. The mesh size increase ratio was approximately 1.1 from the inside to the outside. To assess the reliability of our numerical simulation, we conducted a comparison of drag coefficients with experimental results obtained by Liu et al. [22] for the same parallel twin box girders with corresponding gap distance ratios. As demonstrated in Fig. 14, the results exhibit good concordance, suggesting that our numerical simulation precisely replicated the flow field around the single box girder and

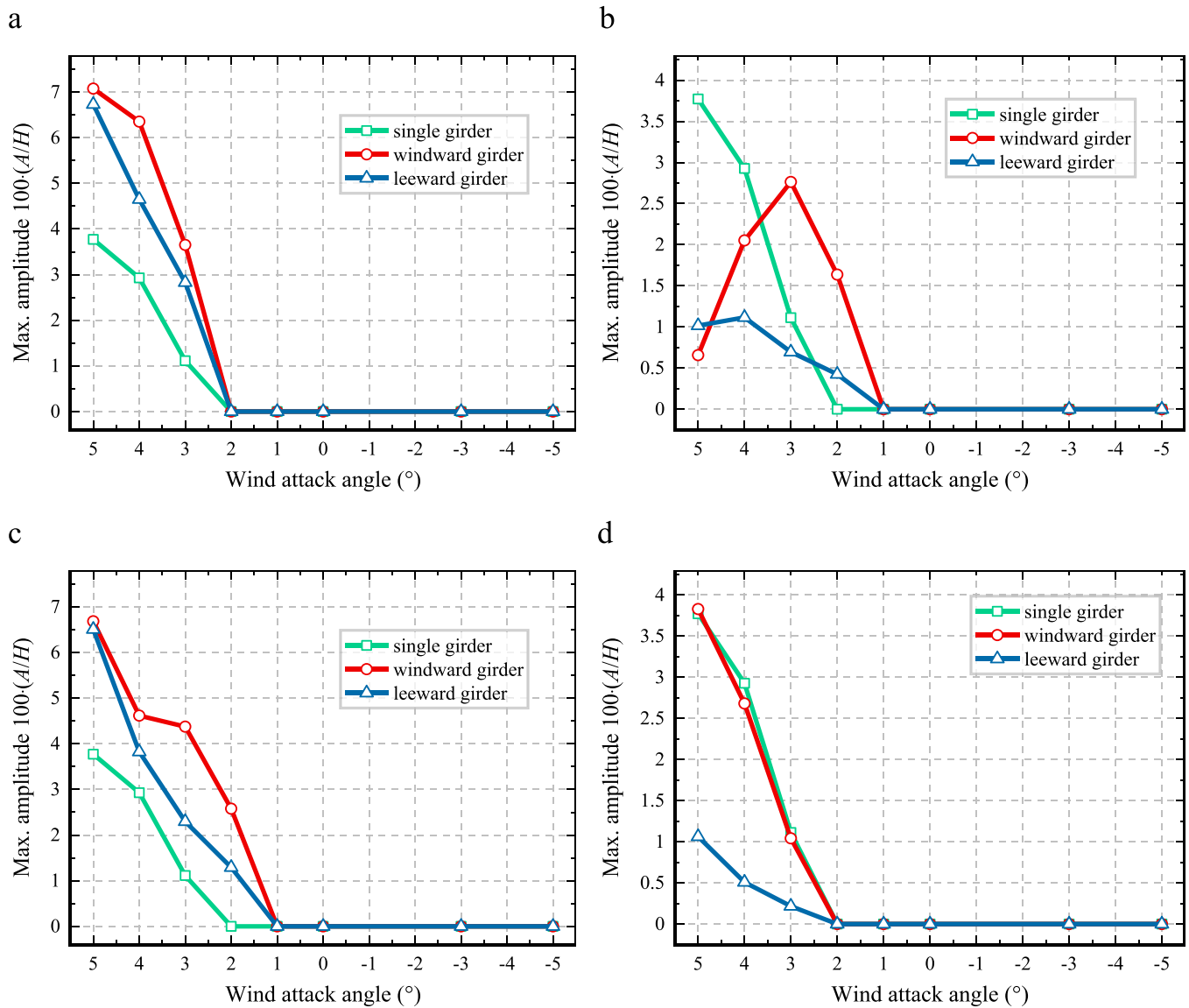


Fig. 12. Maximum VIV amplitude with AoA of the single box girder and twin box girders at gap distances of (a) $D/B = 0.025$, (b) $D/B = 0.1$, (c) $D/B = 0.4$ and (d) $D/B = 3.0$.

twin box girders for each gap distance. Our results provide a basis for the mechanism analysis of the aerodynamic interference in sectional model vibration tests.

4.2. Numerical simulation results

Fig. 15(a) and (b) display the time-averaged streamlines and TKE fields for the single box girder and twin box girders at different gap distances, respectively. The physical significance of TKE is the measure of the turbulence in fluid as it passes through a point. The TKE field can reveal the intensity of vortex shedding in the flow field. In the following analysis, the mechanism of flow field aerodynamic interference between parallel twin box girders at different gap distances is examined through a comparison with a single box girder.

When the incoming flow passes through the single box girder, flow separation occurs at the windward side of the upper surface. Subsequently, a large-scale vortex with high intensity forms on the upper surface of the single box girder, which causes girder VIV to occur at certain wind speeds.

At a minimal gap distance ($D/B \approx 0.025$), large-scale vortices are present on the upper surfaces of both the windward and leeward box

girders. The intensity of vortex shedding on the upper surface of the windward box girder is similar to that of the single box girder. Additionally, there is alternate vortex shedding between the inclined webs of the box girders. This shedding leads to more adverse flow field conditions for the twin box girders at this gap distance than that for the single box girder, leading to an amplification of VIV.

At a small gap distance ($D/B \approx 0.1$), the intensity of vortex shedding on the upper surfaces of both windward and leeward box girders are lower than that of the single box girder. Furthermore, the leeward box girder effectively reduces the level of vortex shedding in the wake of the windward box girder, which may be responsible for the decrease in the VIV of twin box girders.

At a medium gap distance ($D/B \approx 0.4$), vortices of comparable size to a single box girder manifest on the upper surfaces of both windward and leeward box girders. Moreover, the increased gap distance between box girders not only eliminates the reduction effect of the leeward box girder on vortex shedding in the wake of the windward box girder but also results in prominent vortex shedding between twin box girders. This phenomenon results in the leeward box girder being subjected to the effect of the vortex shedding from both itself and the windward box girder wake. Furthermore, the significant VIV of the leeward box girder

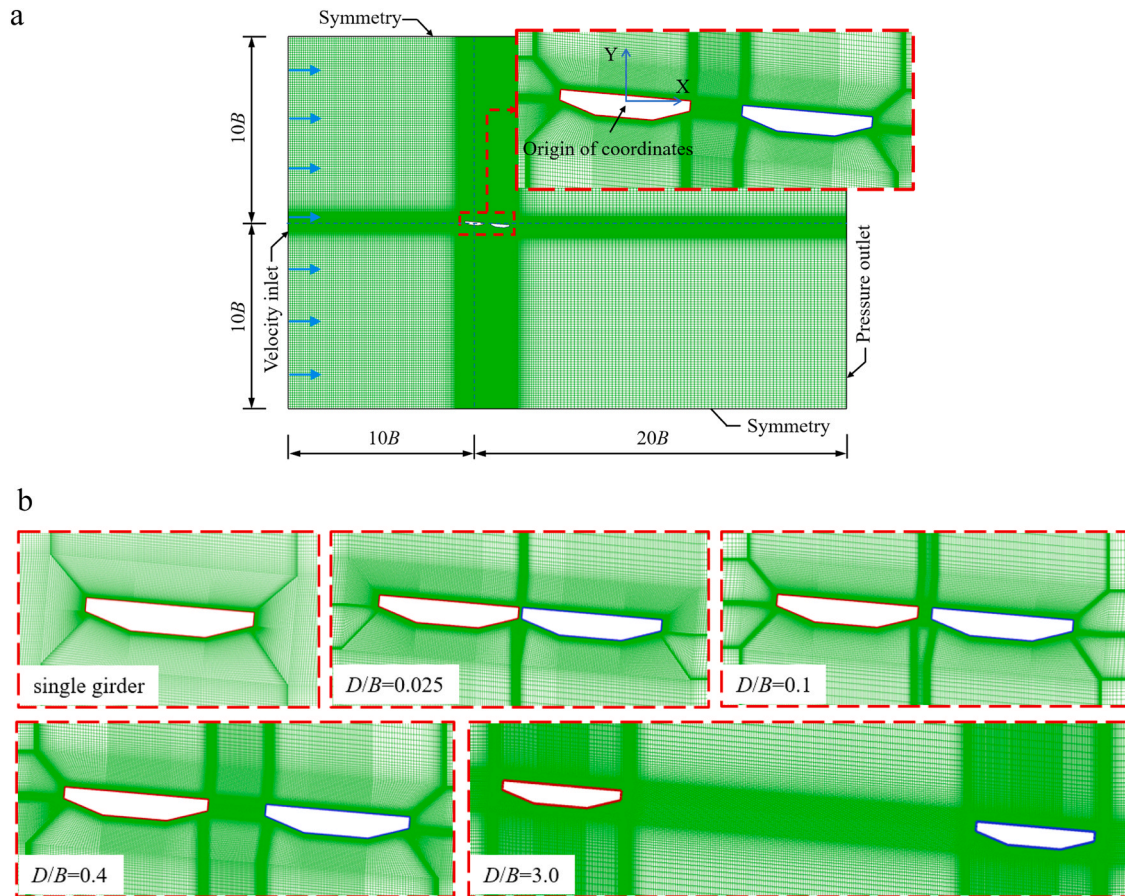


Fig. 13. CFD simulation of 2D flow field: (a) setup of computing domain and overall mesh generation and (b) local amplifications of the single box girder and twin box girders at different gap distances.

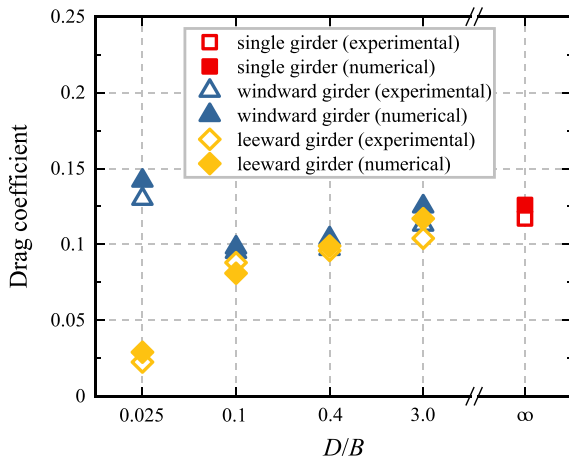


Fig. 14. Comparison of numerical simulation results with experimental results based on drag coefficients.

affects the windward box girder. Thus, the VIV characteristics of twin box girders are amplified relative to the single box girder.

At a large gap distance ($D/B \approx 3.0$), both the windward and leeward box girders exhibit nearly identical vortex sizes and vortex shedding intensities as the single box girder due to the large gap distance. Additionally, since the leeward box girder is situated on the downwind side, it has a minimal effect on the flow field of the windward box girder. Consequently, the VIV characteristics of the windward box girder at a large gap distance closely resemble those of the single box girder. In

contrast, the leeward box girder is exposed to the turbulent field generated by the wake flow of the windward box girder. Since turbulent flow has a reduction effect on the VIV, it may explain why the VIV of the leeward box girder decreases at a large gap distance. It can be anticipated that as the gap distance further increases and the turbulence intensity of the wake flow of the windward box girder decreases, this reduction effect gradually weakens. Consequently, the VIV characteristics of the leeward box girder should become more similar to those of the single box girder than before.

5. Conclusions

We meticulously investigated the vortex-induced vibration (VIV) characteristics of parallel twin steel box girders through sectional model vibration wind tunnel tests. This investigation was performed at various gap distance ratios (D/B , i.e., the ratio of the gap distance (D) of two box girders to the breadth (B) of a single box girder) ranging from 0.025 to 4.0 and initially at an angle of attack (AoA) of $+5^\circ$. The investigation proceeded with four representative gap distances at AoAs ranging from -5° to $+5^\circ$. The results above were compared with those obtained from the single box girder. Finally, we applied the CFD simulation approach to elucidate the mechanism of VIV characteristics at different gap distances and AoAs. The following results were obtained:

(1) The VIV characteristics of twin box girders at AoAs ranging from $+3^\circ$ to $+5^\circ$ can be classified into four intervals based on gap distances: minimum, small, medium, and large gap distances. The ranges of the intervals vary with the AoA. At an AoA of $+5^\circ$, the windward box girder exhibits minimal, small, medium, and large gap distance ranges of $0.025 < D/B < 0.075$, $0.075 < D/B < 0.2$, $0.2 < D/B < 1.5$, and $1.5 < D/B < 4.0$, respectively. The leeward box girder exhibits minimal, small,

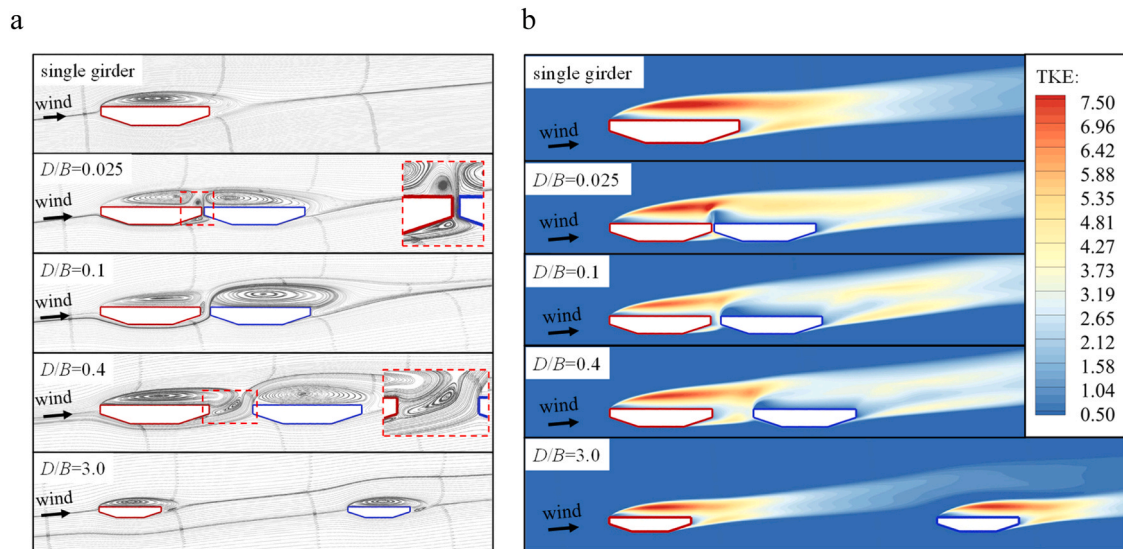


Fig. 15. (a) Time-averaged streamlines and (b) TKE fields for the single box girder and parallel twin box girders at varying gap distances and an AoA of $+5^\circ$.

medium, and large gap distance ranges of $0.025 < D/B < 0.075$, $0.075 < D/B < 0.25$, $0.25 < D/B < 2.5$, and $2.5 < D/B < 4.0$, respectively.

(2) At a minimal gap distance, an amplification effect occurs on the VIV of the twin box girders. This effect is expressed as an amplified amplitude and an extension of the lock-in region. The occurrence of alternating vortex shedding between the inclined webs of the box girders can explain this phenomenon.

(3) At a small gap distance, the VIV of the twin box girders decreases. This reduction is evident by decreases in the VIV amplitude and the lock-in region length. One possible reason for this phenomenon is the presence of the leeward box girder that partially inhibits vortex shedding in the wake of the windward box girder.

(4) At a medium gap distance, there is an amplification effect on the VIV response of twin box girders, as evidenced by the increased VIV amplitude and expansion of the lock-in region length. This phenomenon can be explained by vortex shedding between the windward and leeward box girders.

(5) At a large gap distance, the flow field around the windward box girder is virtually identical to that around the single box girder. Therefore, the VIV characteristics of the windward box girder are similar to those of the single box girder. However, the leeward box girder is situated in the turbulent flow field generated by the wake of the windward box girder, resulting in the reduction effect of the VIV of the leeward box girder. The VIV amplitude of the leeward box girder is smaller than that of the single box girder, and the length of the lock-in region is approximately the same as that of the single box girder.

CRediT authorship contribution statement

Qun Yang: Funding acquisition. **Dewang Zhang:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Anjie Chen:** Software, Resources. **Haidong Zhang:** Data curation, Investigation. **Xiaobing Liu:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Álvarez AJ, Nieto F, Kwok KCS, Hernández S. A computational study on the aerodynamics of a twin-box bridge with a focus on the spanwise features. *J Wind Eng Ind Aerodyn* 2021;209:104465.
- [2] Argentini T, Rocchi D, Zasso A. Aerodynamic interference and vortex-induced vibrations on parallel bridges: the ewijk bridge during different stages of refurbishment. *J Wind Eng Ind Aerodyn* 2015;147:276–82.
- [3] Chen W-L, Li H, Hu H. An experimental study on the unsteady vortices and turbulent flow structures around twin-box-girder bridge deck models with different gap ratios. *J Wind Eng Ind Aerodyn* 2014;132:27–36.
- [4] Chen X, Ma C, Xian R, Liu T. Effects of gap configuration on vortex-induced vibration of twin-box girder and countermeasure study. *J Wind Eng Ind Aerodyn* 2022;231:105232.
- [5] Chen Z, Niu H, Liu Z. Study of aerodynamic interference effect on main girders of Twin-Deck bridges. *Bridge Constr* 2007;9–12.
- [6] de Miranda S, Patruno L, Ricci M, Ubertini F. Numerical study of a twin box bridge deck with increasing gap ratio by using RANS and LES approaches. *Eng Struct* 2015;99:546–58.
- [7] Gao D, Deng Z, Yang W, Chen W. Review of the excitation mechanism and aerodynamic flow control of vortex-induced vibration of the main girder for long-span bridges: a vortex-dynamics approach. *J Fluids Struct* 2021;105:103348.
- [8] Ge B, Ma R, Li F, Hu X, Chen A. Probabilistic vortex-induced vibration occurrence prediction of the twin-box girder for long-span cable-stayed bridges based on wind tunnel tests. *Eng Struct* 2022;262:114325.
- [9] Guo Z, Meng X, Zhou Q, Zhu L. Aerodynamic interference effects of an existed bridge on aerodynamic coefficients of an adjacent new bridge. *Eng Mech* 2010;27:181–186+200.
- [10] He X, Kang X, Yan L, Flay RGJ, Ren P, Wu T. Numerical investigation of flow structures and aerodynamic interference around stationary parallel box girders. *J Wind Eng Ind Aerodyn* 2021;215:104610.
- [11] Irwin PA, Stoyanoff S, Xie J, Hunter M. Tacoma narrows 50 years later—wind engineering investigations for parallel bridges. *Bridge Struct* 2005;1:3–17.
- [12] JTG/T 3360-01-2018. Wind-Resistant Design Specification For Highway Bridges. Beijing, China: China Communications Press Co., Ltd; 2018.
- [13] Kim S-J, Kim H-K, Calmer R, Park J, Kim GS, Lee DK. Operational field monitoring of interactive vortex-induced vibrations between two parallel cable-stayed bridges. *J Wind Eng Ind Aerodyn* 2013;123:143–54.
- [14] Kimura K, Shima K, Sano K, Kubo Y, Kato K, Ukon H. Effects of separation distance on wind-induced response of parallel box girders. *J Wind Eng Ind Aerodyn* 2008;96:954–62.
- [15] Laima S, Jiang C, Li H, Chen W, Ou J. A numerical investigation of reynolds number sensitivity of flow characteristics around a twin-box girder. *J Wind Eng Ind Aerodyn* 2018;172:298–316.

- [16] Laima S, Li H, Chen W, Li F. Investigation and control of vortex-induced vibration of twin box girders. *J Fluids Struct* 2013;39:205–21.
- [17] Laima S, Li H. Effects of gap width on flow motions around twin-box girders and vortex-induced vibrations. *J Wind Eng Ind Aerodyn* 2015;139:37–49.
- [18] Laima S, Li H, Chen W, Ou J. Effects of attachments on aerodynamic characteristics and vortex-induced vibration of twin-box girder. *J Fluids Struct* 2018;77:115–33.
- [19] Lee S, Kwon S-D, Yoon J. Reynolds number sensitivity to aerodynamic forces of twin box bridge girder. *J Wind Eng Ind Aerodyn* 2014;127:59–68.
- [20] Li H, Laima S, Jing H. Reynolds number effects on aerodynamic characteristics and vortex-induced vibration of a twin-box girder. *J Fluids Struct* 2014;50:358–75.
- [21] Liu Z, Chen Z, Liu G, Shao X. Experimental study of aerodynamic interference effects on aerostatic coefficients of twin deck bridges. *Front Archit Civ Eng China* 2009;3:292–8.
- [22] Liu X, Li S, Yang Q, Liu Q. Influence of aerodynamic interference effect on drag coefficient of twin parallel box girders. *China J Highw Transp* 2017;30:108–14.
- [23] Liu S, Zhao L, Fang G, Hu C, Ge Y. Investigation on aerodynamic force nonlinear evolution for a central-slotted box girder under torsional vortex-induced vibration. *J Fluids Struct* 2021;106:103380.
- [24] Liu L, Zou Y, He X, Yang J, Wang Z, Guo D. Experimental investigation on vortex-induced vibration of a long-span rail-cum-road bridge with twin separated parallel decks. *J Wind Eng Ind Aerodyn* 2022;228:105086.
- [25] Ma C, Pei C, Liao H-l, Liu M, Li M. Field measurement and wind tunnel study of aerodynamic characteristics of twin-box girder. *J Wind Eng Ind Aerodyn* 2020;202: 104209.
- [26] Mashnad M, Jones NP. A model for vortex-induced vibration analysis of long-span bridges. *J Wind Eng Ind Aerodyn* 2014;134:96–108.
- [27] Meng X, Zhu L, Guo Z. Aerodynamic interference effects and mitigation measures on vortex-induced vibrations of two adjacent cable-stayed bridges. *Front Archit Civ Eng China* 2011;5:510–7.
- [28] Park J, Kim H-K. Effect of the relative differences in the natural frequencies of parallel cable-stayed bridges during interactive vortex-induced vibration. *J Wind Eng Ind Aerodyn* 2017;171:330–41.
- [29] Park J, Kim S, Kim H-K. Effect of gap distance on vortex-induced vibration in two parallel cable-stayed bridges. *J Wind Eng Ind Aerodyn* 2017;162:35–44.
- [30] Sato H, Hirahara N, Fumoto K, Hirano S, Kusuhara S. Full aeroelastic model test of a super long-span bridge with slotted box girder. *J Wind Eng Ind Aerodyn* 2002;90: 2023–32.
- [31] Sato H, Kusuhara S, Ogi K-i, Matsufuji H. Aerodynamic characteristics of super long-span bridges with slotted box girder. *J Wind Eng Ind Aerodyn* 2000;88: 297–306.
- [32] Seo J-W, Kim H-K, Park J, Kim K-T, Kim G-N. Interference effect on vortex-induced vibration in a parallel twin cable-stayed bridge. *J Wind Eng Ind Aerodyn* 2013;116: 7–20.
- [33] Tang H, Shum KM, Li Y. Investigation of flutter performance of a twin-box bridge girder at large angles of attack. *J Wind Eng Ind Aerodyn* 2019;186:192–203.
- [34] Wu T, Kareem A. An overview of vortex-induced vibration (VIV) of bridge decks. *Front Struct Civ Eng* 2012;6:335–47.
- [35] Wu T, Kareem A. Vortex-Induced vibration of bridge decks: volterra Series-Based model. *J Eng Mech* 2013;139:1831–43.
- [36] Wu B, Laima S. Experimental study on characteristics of vortex-induced vibration of a twin-box girder and damping effects. *J Fluids Struct* 2021;103:103282.
- [37] Xu K, Bi K, Han Q, Li X, Du X. Using tuned mass damper inerter to mitigate vortex-induced vibration of long-span bridges: analytical study. *Eng Struct* 2019;182: 101–11.
- [38] Yang Y, Zhang L, Ding Q, Ge Y. Flutter performance and improvement for a suspension bridge with central-slotted box girder during erection. *J Wind Eng Ind Aerodyn* 2018;179:118–24.
- [39] Yang Y, Zhou R, Ge Y, Mohotti D, Mendis P. Aerodynamic instability performance of twin box girders for long-span bridges. *J Wind Eng Ind Aerodyn* 2015;145: 196–208.
- [40] Yang Y, Zhou R, Ge Y, Zhang L. Experimental studies on VIV performance and countermeasures for twin-box girder bridges with various slot width ratios. *J Fluids Struct* 2016;66:476–89.
- [41] Yang Y, Zhou R, Ge Y, Zou X, Zhang L. Flutter characteristics of twin-box girder bridges with vertical central stabilizers. *Eng Struct* 2017;133:33–48.
- [42] Zhou R, Ge Y, Yang Y, Du Y, Liu S, Zhang L. Nonlinear behaviors of the flutter occurrences for a twin-box girder bridge with passive countermeasures. *J Sound Vib* 2019;447:221–35.
- [43] Zhou R, Ge Y, Yang Y, Liu S, Du Y, Zhang L. A nonlinear numerical scheme to simulate multiple wind effects on twin-box girder suspension bridges. *Eng Struct* 2019;183:1072–90.
- [44] Zhou R, Yang Y, Ge Y, Zhang L. Comprehensive evaluation of aerodynamic performance of twin-box girder bridges with vertical stabilizers. *J Wind Eng Ind Aerodyn* 2018;175:317–27.
- [45] Zhu Q, Xu YL, Shum KM. Stress-level buffeting analysis of a long-span cable-stayed bridge with a twin-box deck under distributed wind loads. *Eng Struct* 2016;127: 416–33.